

Cell Design Specification — VBF-T10A-DS-01

Case: next-generation pure electric sedan battery. Values mechanically taken from parameter set / simulation output; sources annotated per line. Missing items marked "Not provided".

1. Basic specification

Field	Value	Source
Electrochemical system	NMC811 positive / graphite+SiOx negative (OKane2022)	parameter set `--base OKane2022`
Nominal capacity	5.0 Ah (nominal) / 5.057 Ah (1C DFN simulated)	parameter set `cell/T10_1c_dfn.json`
Voltage window	2.5 – 4.2 V	parameter set (Lower/Upper voltage cut-off)
Cell dimensions (h × w × t)	65 mm × 1580 mm × 200.8 μm (active stack; shell thickness Not provided)	parameter set geometry + `cell/T10_energy_dfn.json:thickness_m`
Electrolyte formulation	advanced formulation: conductivity 3.0 S/m, transference number $t_{+}=0.6$, diffusivity 4.0e-10 m²/s (estimate, advanced LiFSI/carbonate-class)	design override `params_T10.json`
Cation transference number	0.6	design override
Thermal management	liquid cooling, h = 80 W/m²/K	design override

2. Electrode and separator

Layer	Thickness (μm)	Porosity	Active-material vol. frac.	Current collector
Positive (NMC811)	75.6	0.335	0.665	Al, 16 μm
Separator (polyolefin)	12	0.47	—	—
Negative (graphite+SiOx)	85.2	0.25	0.75	Cu, 12 μm

• N/P ratio = (negative areal Li capacity) ÷ (positive areal Li capacity) = $(0.75 \times 85.2 \mu\text{m} \times 33133 \text{ mol/m}^3) \div (0.665 \times 75.6 \mu\text{m} \times 63104 \text{ mol/m}^3) = 2.118 \div 3.172 = \text{**}0.667\text{**}$. Note: computed from the parameter set's maximum concentrations (graphite-basis for the negative); the SiOx component contributes additional capacity via its own OCP, and plating safety at 4C is verified (+0.0341 V) independent of this c_max-based ratio.

3. Process design parameters

Parameter	Positive	Negative	Source
Areal density (g/m²)	163.99	105.88	thickness × (1–porosity) × density
Compaction density (g/cm³)	2.169	1.243	density × (1–porosity) ÷ 1000
Electrolyte fill amount	6.44 g/cell (total pore vol. 5.37 cm³ × 1.2 g/cm³)	—	pore volume × 1.2 g/cm³ (literature value)
Formation recommendation	0.1C CC to 4.2 V, 25 °C, 2 cycles	—	design recommended value; production value requires tuning

4. Mass breakdown

Layer	kg/m²	g/cell	Source
Positive electrode	0.16399	16.84	`cell/T10_energy_dfn.json:layer_kg_m2` × area 0.1027 m²
Negative electrode	0.10588	10.87	same
Positive current collector (Al)	0.04320	4.44	same
Negative current collector (Cu)	0.10752	11.04	same
Separator	0.002525	0.259	same
Total (electrolyte excluded)	0.42312	**43.45**	`cell/T10_energy_dfn.json:mass_kg`

Note: contract-caliber mass excludes electrolyte (parameter set lacks electrolyte density) and casing.

5. Performance verification

Item	Condition	Result	Determination	Source
1C discharge capacity	1C, 298 K	5.057 Ah	✓ (nominal 5.0)	`cell/T10_1c_dfn.json`
Energy density	contract caliber	426.89 Wh/kg	✓ (≥ 392.61)	`cell/T10_energy_dfn.json`
Volumetric energy density	contract caliber	899.5 Wh/L	—	`cell/T10_energy_dfn.json`
4C fast charge, plating	4C, 318 K, plating module	anode min +0.0341 V $\rightarrow$ no plating	✓ (plated=false)	`cell/T10_4c_dfn.json`
4C temperature rise	4C, 318 K, lumped thermal	T_max 324.55 K (51.4 °C)	✓ (≤ 333.15 K)	`cell/T10_4c_dfn.json`
Overcharge 4.7 V thermal runaway	0.5C charge to 4.7 V + run-tr	triggered = false	✓	`cell/T10_tr.json`

6. Design notes

Changed vs OKane2022 baseline: (1) cooling h 10 $\rightarrow$ 80 W/m²/K to cap 4C temperature (368 $\rightarrow$ 324.55 K); (2) electrolyte transport σ 1.x $\rightarrow$ 3.0 S/m, t 0.259 $\rightarrow$ 0.6, D $\rightarrow$ 4e-10 m²/s; (3) negative particle radius 5.86 $\rightarrow$ 0.8 μ m (nanostructured SiOx) and positive 5.22 $\rightarrow$ 2.0 μ m. Rationale (evaluate log R2–R4): graphite anode (Chen2020) plates fundamentally at 4C (anode -0.438 V, transport-insensitive); SiOx anode does not plate but overheats at default cooling; liquid cooling caps T_max while advanced transport + nanostructured anode restore the no-plating margin at the cooler operating temperature.